



Bulacan State University
City of Malolos, Bulacan



College of Information and Communications Technology
Information Technology Department

Library Management System

Project Title:

FINAL PROJECT

in

IT 203

OBJECT-ORIENTED PROGRAMMING 1

Submitted by:

DIMLA, Tyron B.

MENDOZA, Reymark M.

OXALES, Emmanuel D.G.

QUIÑONES JR, Romeo M.

TAGANAS, Lorenz Romeo O.

BSIT 2G – G1

Submitted to:

ANIAG, Jennylyn L.

Library Management System

Our Library Management System (LMS) is a comprehensive and user-friendly platform designed to streamline the management of library resources and facilitate easy access for students, teachers, staff, and librarians. The system is built with a user-centric approach, ensuring a seamless experience for all users.

Key Functionalities:

Sign Up:

- ❖ **New Member Registration:** Prospective users, including students, teachers, and staff, can easily register for library access by providing essential details such as name, email, and a unique username/password combination.

Login:

- ❖ **Secure Access:** Registered members can log in securely using their unique username and password, ensuring that only authorized individuals can access their library accounts.
- ❖ **Dashboard Access:** Upon successful login, users are directed to a personalized dashboard where they can explore the library catalog, manage borrowed books, and access other relevant features.

Librarian:

- ❖ **Manage Books:** Librarians can efficiently manage the library's collection, including adding new books, updating existing book information, and deleting/removing book records.
- ❖ **Manage Members:** Librarians have the ability to manage the library's user database. This includes adding new members, updating member information, and deleting or removing members when necessary.
- ❖ **Manage Borrowed Books:** Librarians can oversee and track borrowed books, handle book returns..

Student/Teacher/Staff:

- ❖ **View Books:** Users can browse the library catalog, search for specific books, and view detailed information about each book, including availability status.
- ❖ **Borrow Books:** Members can easily borrow books by double-clicking on a specific book. After clicking the "Borrow" option, the borrowing transaction will be stored in the database.
- ❖ **View Borrowed Books:** Users have access to a personalized dashboard where they can view a list of books they have borrowed, check due dates.

Key Features:

Search and Filters:

- ❖ **Search and Filters:** A robust search functionality allows users to quickly find books based on titles, authors, genres, or keywords. Filters enable precise searches, making it convenient for users to locate specific materials.
- ❖ **User-Friendly Interface:** The system boasts an intuitive and user-friendly interface, ensuring a smooth navigation experience for both librarians and library members.

Benefits:

- ❖ **Efficiency:** Streamlines library operations, reducing manual effort in book management.
- ❖ **Accessibility:** Enhances accessibility to the library's resources for all users, promoting a culture of learning.
- ❖ **Transparency:** Provides transparency in borrowing processes, allowing users to monitor their activity and librarians to track library usage.

Our Library Management System is a powerful tool designed to create a harmonious environment for both librarians and library users, fostering a culture of knowledge sharing and exploration.

Log in/Sing up

Sign up

Library Management System

Sign up

Sign up as...

Select...

First Name

Last Name

Email

Username

Password

show password

SIGNUP

Already have an account? [Log in](#)

Sign up

Library Management System

Sign up

Sign up as...

Select...

Librarian

Teacher

Student

Staff

Username

Password

show password

SIGNUP

Already have an account? [Log in](#)

Our Library Management System offers a user-friendly **Sign Up** feature, allowing individuals to register. Users can sign up based on their roles, choosing from options such as Librarian, Teacher, Student or Staff.

Login

Library Management System

Log in

Log in as...

Select...

Username

Password

show password

LOGIN

Don't have an account? [Sign up](#)

Login

Library Management System

Log in

Log in as...

Select...

Librarian

Teacher

Student

Staff

Password

show password

LOGIN

Don't have an account? [Sign up](#)

Our Library Management System provides a straightforward and secure **Log in** feature, ensuring seamless access for users based on their roles: Librarian, Student, Teacher, or Staff.

Librarian

Manage Books

Welcome back!

Jennylyn

Manage Books

Manage Members

Manage Borrowed Books

Log out

Search

Filter Books byCategory...

ISBN	Title	Author	Publish Date	Category	Status	Quantity
9780134494166	Introduction to ...	Thomas H. Cor...	2009-07-31	Programming	Not Available	0
9780374533557	Thinking, Fast a...	Daniel Kahnem...	2011-10-25	Science	Not Available	0
9780446520706	The Catcher in t...	J.D. Salinger	1951-07-16	English	Not Available	0
9780465026562	Gödel, Escher, B...	Douglas Hofsta...	1979-04-01	Philosophy	Available	0
9780521294149	Principia Mathe...	Alfred North	1910-01-01	Math	Not Available	1
9781234567890	Code Craf...	...	2022-05-15	Programming	Available	2
9781234567891	The Quar...	...	...	Science	Available	2
9781234567892	Mathema...	...	...	Math	Available	7
9781234567893	Language...	...	...	English	Not Available	0
9781234567896	The Cosm...	...	...	Science	Available	2
9781234567897	Quantum...	...	...	Math	Available	2
9781234567898	Mastering...	...	...	English	Not Available	0
9781234567899	Existential...	...	...	Philosophy	Available	3

Title:Introduction to Algorithms

Author:Thomas H. Cormen

ISBN:9780134494166

Publish Date:2009, 07, 31

Status:Not Available

Quantity:0

Category:Programming

CLEAR

ADD

UPDATE

DELETE

Book Details

Title: Introduction to Algorithms

Author: Thomas H. Cormen

ISBN: 9780134494166

Publish Date: 2009-07-31

Category: Programming

Status: Not Available

Quantity: 0

OK

Log out

On the “**Manage Books**”, the role of the librarian encompasses efficient control and organization of the library's collection. Librarians have the capability to **add, update, and delete book records**, ensuring a well-maintained and up-to-date catalog for the library's users.

Manage Members

Welcome Back!

Jennylyn

Manage Books

Manage Members

Manage Borrowed Books

Log out

Search:

Filter Members by:Role...

User_ID	First Name	Last Name	Email	Username	Password	Role
1	Jennylyn	Aniag	jennylynaniag@...	admin	password	Librarian
2	Romeo	Quinones	romeoquinones...	2022100960	romeo960	Student
3	Tyron	Dimla	tyrondimla@gm...	2022100499	tyron499	Student
4	Emman	Oxales	emmanoxales@...	emman123	emman321	Student
6	Reymark	Mendoza	reymarkmendoz...	mendoza123	mendoza321	Teacher
7	Lorenz Romeo	Taganas	lorenzromeotag...	lorenzromeo123	lorenzromeo321	Staff

First Name:Jennylyn

Last Name:Aniag

CLEAR

Email:jennylynaniag@gmail.com

ADD

Username:admin

UPDATE

Password:password

DELETE

Role:Librarian

Log out

On the “**Manage Members**” empowers librarians with comprehensive tools to oversee and maintain member records within the Library Management System. **Librarians have the capability to add, update, and delete member records.**

Manage Borrowed Books

Welcome back!

Jennylyn

Manage Books

Manage Members

Manage Borrowed Books

Log out

Search

Filter Books byCategory...

Borrow_ID	Title	Category	Borrower First N...	Borrower Last N...	Borrowed Date	Return Date
1	Introduction to ...	Programming	Romeo	Quinones	2023-11-28	2023-12-13
8	Introduction to ...	Programming	Reymark	Mendoza	2023-11-28	2023-12-13
17	Introduction to ...	Programming	Lorenz Romeo	Taganas	2023-11-28	2023-12-13
22	Introduction to ...	Programming	Tyron	Dimla	2023-11-28	2023-12-13
25	Introduction to ...	Programming	Emman	Oxales	2023-11-28	2023-12-13
3	Thinking, Fast a...	Science	Quinones	Romeo	2023-11-28	2023-12-13
9	Thinking, Fast a...	Science	Reymark	Mendoza	2023-11-28	2023-12-13
7	Code Craf...	Programming	Quinones	Romeo	2023-11-28	2023-12-13
16	The Qua...	Math	Taganas	Mendoza	2023-11-28	2023-12-13
19	The Cos...	Math	Taganas	Quinones	2023-11-28	2023-12-13
24	Mathem...	Math	Dimla	Mendoza	2023-11-28	2023-12-13
12	Mathem...	Math	Taganas	Oxales	2023-11-28	2023-12-13
20	Mathem...	Math	Mendoza	Quinones	2023-11-28	2023-12-13
6	The Cos...	Math	Quinones	Mendoza	2023-11-28	2023-12-13
10	Quantum Mathe...	Math	Dimla	Quinones	2023-11-28	2023-12-13
23	Mastering Engli...	English	Romeo	Quinones	2023-11-28	2023-12-13
5	Existential Musi...	Philosophy	Romeo	Quinones	2023-11-28	2023-12-13
13	Existential Musi...	Philosophy	Reymark	Mendoza	2023-11-28	2023-12-13
21	Existential Musi...	Philosophy	Tyron	Dimla	2023-11-28	2023-12-13
27	Existential Musi...	Philosophy	Emman	Oxales	2023-11-28	2023-12-13

Borrow_ID: 1

Title: Introduction to Algorithms

Category: Programming

Borrower Name: Romeo Quinones

Borrowed Date: 2023-11-28

Return Date: 2023-12-13

Returned

Close

Returning Book Confirmation

Has this book already been returned?

Yes

No

Log out

Manage Borrowed Books

Welcome back!

Jennylyn

Manage Books

Manage Members

Manage Borrowed Books

Log out

Search

Filter Books byCategory...

Borrow_ID	Title	Category	Borrower First N...	Borrower Last N...	Borrowed Date	Return Date
1	Introduction to ...	Programming	Romeo	Quinones	2023-11-28	2023-12-13
8	Introduction to ...	Programming	Reymark	Mendoza	2023-11-28	2023-12-13
17	Introduction to ...	Programming	Lorenz Romeo	Taganas	2023-11-28	2023-12-13
22	Introduction to ...	Programming	Tyron	Dimla	2023-11-28	2023-12-13
25	Introduction to ...	Programming	Emman	Oxales	2023-11-28	2023-12-13
3	Thinking, Fast a...	Science	Romeo	Quinones	2023-11-28	2023-12-13
9	Thinking, Fast a...	Science	Reymark	Mendoza	2023-11-28	2023-12-13
7	Code Craftsman...	Programming	Romeo	Quinones	2023-11-28	2023-12-13
16	The Qua...	Math	Taganas	Mendoza	2023-11-28	2023-12-13
19	The Cos...	Math	Taganas	Quinones	2023-11-28	2023-12-13
24	Mathem...	Math	Dimla	Mendoza	2023-11-28	2023-12-13
12	Mathem...	Math	Taganas	Oxales	2023-11-28	2023-12-13
20	Mathem...	Math	Mendoza	Quinones	2023-11-28	2023-12-13
6	The Cos...	Math	Quinones	Mendoza	2023-11-28	2023-12-13
10	Quantum Mathe...	Math	Dimla	Quinones	2023-11-28	2023-12-13
23	Mastering Engli...	English	Romeo	Quinones	2023-11-28	2023-12-13
5	Existential Musi...	Philosophy	Romeo	Quinones	2023-11-28	2023-12-13
13	Existential Musi...	Philosophy	Reymark	Mendoza	2023-11-28	2023-12-13
21	Existential Musi...	Philosophy	Tyron	Dimla	2023-11-28	2023-12-13
27	Existential Musi...	Philosophy	Emman	Oxales	2023-11-28	2023-12-13

Returning Book Confirmation

Has this book already been returned?

Yes

No

Log out

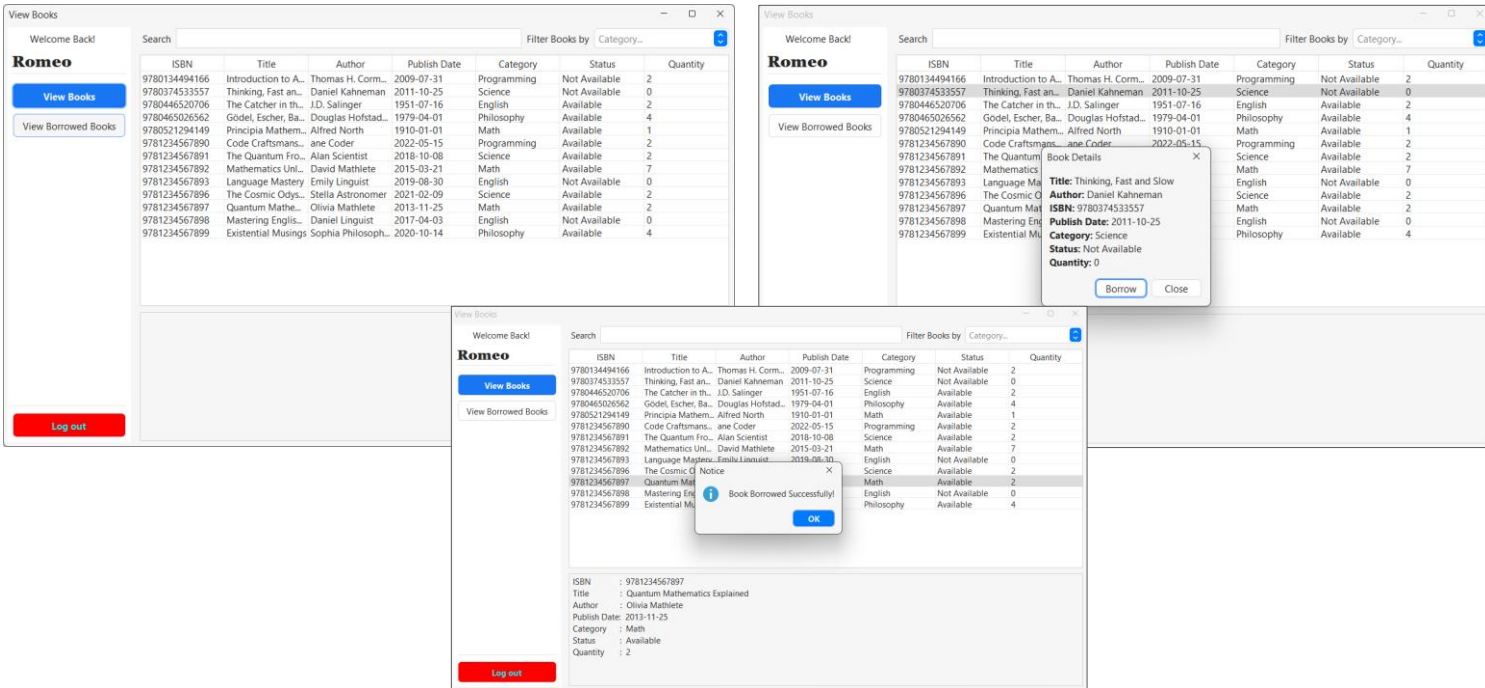
On the “**Manage Borrowed Books**”, **librarians** can click on specific borrowing records within the system to access detailed information about the borrowed book and borrower.

Upon selecting a record, an option dialog promptly appears, presenting the librarian with choices, including the option to mark the book as “**Returned**”.

Choose to mark a book as “Returned” triggers a secondary confirmation dialog designed to verify if the book has indeed been returned. This dialog includes a question: "Has this book already been returned?".

The Librarian can choose “**Yes**” to **confirm that the book has been returned**, initiating the removal of the borrowing record from the database.

User (Teacher, Student, Staff)

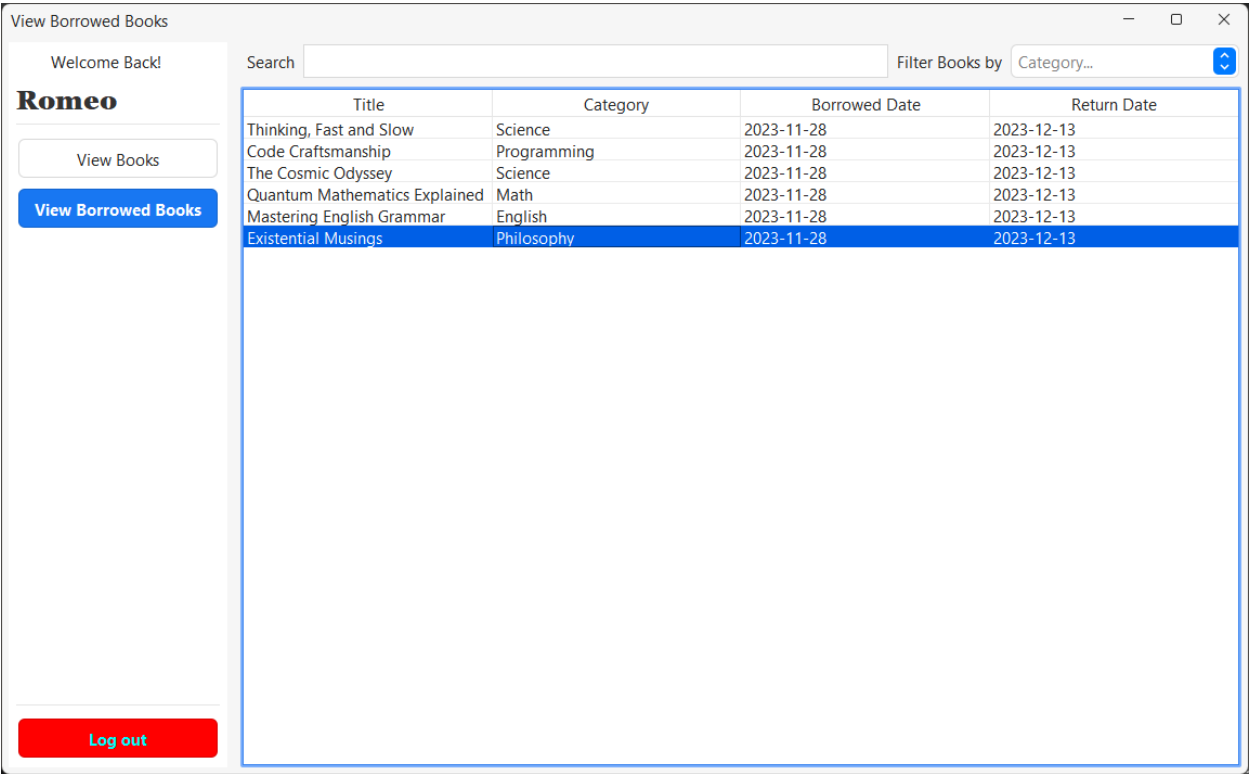


On the “**View Books** ”, **users (Teacher, Student, Staff)** can effortlessly navigate through a user-friendly interface to view a comprehensive list of books categorized based on subjects like Programming, Science, Math, English, and Philosophy.

By clicking on a specific book record, users trigger an interactive option dialog that provides details about the selected book, including the title, author, ISBN, publish date, category, availability status, and quantity.

Upon deciding to borrow a book, users can choose the "Borrow" option from the interactive dialog. A message dialog promptly appears, confirming the successful borrowing of the selected book.

The system dynamically stores the borrowing transaction in the database, associating it with the borrower's unique ID. This ensures accurate and real-time tracking of borrowed books per user.



For Users, the **"View Borrowed Books"** feature provides a convenient way to manage and monitor their borrowed book transactions within the Library Management System.

Users can access comprehensive details for each borrowed book, including title, category, and borrowing date. **The return date is set to 15 days** from the borrowing date, offering a clear timeframe for returning the book.

Search and Filter

